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Solvent-mediated contaminant removal from plastic waste using thermodynamic modeling

 Panzheng Zhou,^{†a} Ugochukwu M. Ikegwu,^{†a} Elizaveta Radkevich,^a Euncheol Ra,^a Hadeia Liburd,^b George W. Huber,^{id}^a Victor M. Zavala^a and Reid C. Van Lehn^{id}^{*a,c}

Plastics recycling is hindered by the compositional complexity of plastic waste, which can include numerous polymer components as well as low concentrations of additives and non-intentionally added substances. These latter small-molecule species, which we collectively refer to as contaminants, can harm human health and will build up in recycled plastic causing environmental and downstream processing challenges if not removed. In this work, we present molecular modeling approaches using the COnductor-like Screening MOdel for Real Solvents (COSMO-RS) to guide the selection of solvents that are capable of removing targeted contaminants from plastic waste. By considering the thermodynamic partitioning of contaminant species between a solvent phase and polymer phase, we identify guidelines for solvent selection to promote either the low-temperature extraction of contaminants from plastic waste or the removal of contaminants as part of a dissolution-based plastics recycling process. We present four case studies to illustrate the application of the computational approach to the removal of brominated flame retardants, phthalates, and selected perfluoroalkyl substances, and compare to both literature and newly collected experimental data to illustrate model prediction accuracy. The case studies highlight the capability of the modeling approach to help design recycling processes that explicitly account for contaminant removal, thereby increasing product purity during dissolution-based recycling or facilitating chemical recycling of contaminant-free plastics.

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1. This work proposes computational methods to identify solvents that can remove hazardous contaminants from plastic waste, thereby facilitating the design of sustainable recycling processes that minimize plastic pollution in the environment.
2. The computational methods developed in this work quantify contaminant partitioning into different solvents based on thermodynamically well-defined partition coefficients. The integration of these calculations with previous methods to predict polymer solubility enable contaminant removal from plastic waste, as demonstrated in multiple case studies.
3. This work could be extended to a broader range of harmful contaminants by integrating the thermodynamic methods with machine learning methods, which could also permit analysis of solvent greenness.

Introduction

Increasing interest in reducing the amount of plastic pollution that ends up in landfills or leaking into the environment has motivated the development of numerous plastics recycling processes, such as mechanical recycling, solvent-based recycling,

and chemical recycling (*e.g.*, pyrolysis).^{1–4} While substantial effort has been devoted to understanding how these processes can recycle the polymer resins in plastic waste, less is known about the impact of other plastic waste components on recycling processes.⁵ Most plastic products are made from base polymer resins mixed with a complex blend of materials collectively known as additives. These additives enhance plastic products by making them safer, more durable, and visually appealing.⁶ Additives also reduce production costs and extend the lifespan of plastic materials.⁷ There are many different types of additives, with recent databases including thousands of unique chemical species.^{8–10} For example, plasticizers, such as phthalates and adipates, increase polymer flexibility by reducing intermolecular forces.¹¹ Stabilizers such as UV absor-

^aDepartment of Chemical and Biological Engineering, University of Wisconsin – Madison, Madison, WI, 53706, USA. E-mail: vanlehn@wisc.edu
^bDepartment of Chemical Engineering, Howard University, Washington, DC, 20059, USA

^cDepartment of Chemistry, University of Wisconsin – Madison, Madison, WI, 53706, USA. E-mail: vanlehn@wisc.edu
[†]These authors contributed equally.

bers and antioxidants can prevent degradation caused by heat, light, and oxidation.¹² Flame retardants inhibit combustion by disrupting the burning process.¹³ Fillers and reinforcements, such as glass fibers and calcium carbonate, improve mechanical strength and reduce material costs.¹⁴ Colorants, including inorganic pigments (e.g., titanium dioxide) and organic dyes, provide coloration and opacity.¹⁵ Antimicrobial agents, such as silver nanoparticles, inhibit microbial growth in medical and food packaging applications.¹⁶ Antistatic agents can minimize electrostatic buildup on plastic surfaces.¹⁷ Other additives include lubricants, nucleating agents, impact modifiers, among others. Additives contribute to the substantial compositional complexity in plastic products.¹⁰

Despite their benefits, some additives raise concerns due to their persistence in the environment and potential health impacts, which have led to regulatory restrictions on their usage.¹⁸ The presence of additives can also pose a challenge for plastics recycling processes due to their potential to be unintentionally present in recycling streams, particularly for post-consumer plastics that are mixed with a wide range of different plastic products and other types of waste.^{19,20} Post-consumer waste plastics can also include non-intentionally added substances (NIAS) that migrate into plastic products from the environment or other waste materials.²¹ NIAS also include substances that are present in plastics without intentionally imparting a technical function to the plastic. As such, NIAS include impurities found in starting materials such as monomers, processing aids, colorants and printing inks, cross-linking agents, as well as by-products or intermediates from polymerization reactions.¹⁰ The most common plastics recycling technology, mechanical recycling, does not remove additives or NIAS without specific preprocessing steps, such as washing in surfactant solutions,^{22,23} which can lead to these molecules persisting into the final recycled product raising regulatory and health concerns.^{24–27} Similarly, additives or NIAS can impact the efficiency or products of chemical recycling processes.²⁸ Consequently, we collectively refer to additives and NIAS as “contaminants” for the remainder of this work due to their undesirable presence in recycling streams and need for removal.

Because contaminants in plastic waste streams pose a substantial hindrance to plastics recycling due to industrial standards regulating the maximum concentration of these chemicals in pure polymers, processes are required to remove potential contaminants prior to or during plastics recycling. Example processes for contaminant removal include washing in water or surfactant solutions or solvent extraction in organic solvents.^{22,23} These processes rely on the selection of solvent or processing conditions to preferentially remove contaminants from the polymer matrix or surface prior to recycling and are often incorporated as pre-processing steps that are applied to waste prior to mechanical or chemical recycling. Alternatively, solvent-based recycling methods that involve the dissolution of polymer resins may also lead to the dissolution of contaminants in the solvent phase, where they can be removed by adsorption.^{4,29} For example, Solvent-Targeted

Recovery and Precipitation (STRAPTM) is a solvent-mediated process in which target polymer resins are recovered from multicomponent plastic waste *via* selective dissolution and precipitation.^{30,31} STRAP has been applied to a variety of complex feedstocks^{31–37} and can also lead to the removal of inks which would otherwise be contaminants in the recovered resins.^{32,33} There is thus the potential to combine solvent-mediated contaminant removal steps with polymer recovery steps, thereby intensifying solvent-based plastics recycling processes while improving product quality. Designing any of these solvent-mediated contaminant removal processes is challenging, however, because each category of contaminant encompasses a large family of hundreds of different chemical compounds and compounds can be present in low concentrations, which can hinder the rational selection of solvent and operating conditions.

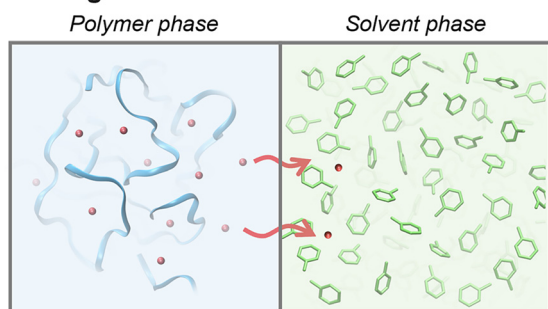
To address this problem, in this work we develop computational modeling methods to screen solvents for selectively removing contaminants from plastic waste. Our approach builds upon our prior computational screening methods for polymer solubility to identify solvents capable of extracting classes of different contaminants without dissolving the target polymer based on thermodynamic calculations.^{38–41} We focus on three categories of contaminants that have recently gained increasing attention due to issues related to toxicity: brominated flame retardants (BFRs), phthalates, and per- and polyfluoroalkyl substances (PFAS). We present strategies for contaminant removal from plastic waste by leveraging computational predictions of contaminant partitioning across a broad range of solvents, supported by experimental validation using real-world applications involving these representative contaminants. Comparison to experiments reveals that thermodynamic modeling is suitable for identifying solvents capable of removing target contaminants through leaching or in combination with STRAP, although kinetic effects may also impact the efficacy of these processes. These advances pave the way for the development of sustainable plastics recycling methods like STRAP that can produce high-quality, contaminant-free plastics, thereby diverting plastic materials from landfills for reuse in place of petroleum-derived virgin plastics.

Methods

Summary of computational screening approach

Fig. 1 sketches computational solvent screening methods for two distinct contaminant removal processes that we refer to as (1) leaching and (2) STRAP contaminant removal. Leaching is a commonly employed method for the removal of undesirable chemicals from industrial products. This process involves the immersion of the targeted polymer into a solvent that facilitates the extraction of dissolved contaminants from the solid polymer matrix.⁴² Leaching is often utilized in environmental remediation and recycling efforts, as it allows for the selective removal of harmful additives that may pose ecological or health risk, ideally without impacting the original target

(a) Leaching



(b) STRAP contaminant removal

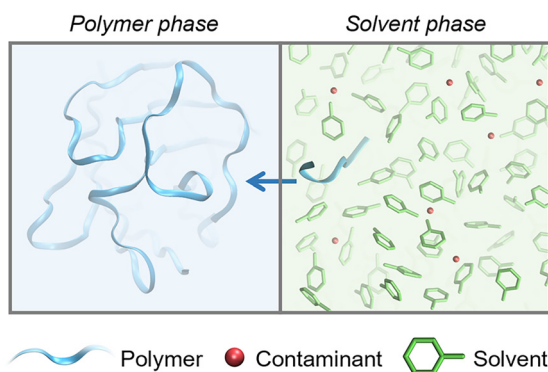


Fig. 1 Molecular-scale illustrations of two distinct contaminant removal strategies. (a) Leaching, where contaminant molecules migrate from the polymer phase into the liquid solvent phase without dissolving the polymer. (b) Simultaneous STRAP and contaminant removal, where the pure polymer is selectively precipitated from a solution containing the dissolved polymer and contaminant.

material. STRAP contaminant removal refers to the incorporation of contaminant removal steps into a dissolution-based plastic recycling process. In the STRAP process, a target polymer is first dissolved in a selective solvent that does not dissolve other polymer components of the waste. The target polymer is then precipitated by decreasing the temperature or adding a nonsolvent to the solution. Therefore, if properly designed, STRAP can be directly used for contaminant removal by dissolving both a target polymer and contaminants then selectively precipitating the polymer without precipitating the dissolved contaminants. The recycling of pure polymer resins and contaminant removal would be achieved in a STRAP process simultaneously. Fig. 1 illustrates both processes for a single contaminant. Our goal is to develop methods that would permit the simultaneous removal of multiple contaminants using either of the two processes shown in Fig. 1.

We hypothesize that the efficiency of contaminant removal depends on the partitioning behavior of contaminant molecules between a solvent phase and an immiscible polymer phase. Such partitioning behavior can be quantified based on the partition coefficient, $\log P$, which is a thermodynamically well-defined parameter that describes the distribution of a solute between the two immiscible liquid phases. Here, we

treat a solid polymer as a liquid phase based on the assumption that contaminants partition into the amorphous region of the polymer matrix with an affinity that can be quantified *via* $\log P$ to guide solvent selection. Based on this assumption, eqn (1) defines the partition coefficient of a single type of contaminant molecule between the two phases, assuming the contaminant is infinitely dilute in each phase.

$$\log P = \log \frac{[c]^s}{[c]^p} = \frac{(\mu_c^p - \mu_c^s)}{RT \ln 10} + \log \left(\frac{V^p}{V^s} \right) \quad (1)$$

Here, c refers to a specific contaminant, the superscript s indicates the solvent, the superscript p indicates the polymer, $[c]^s$ and $[c]^p$ refer to the concentration of contaminant c in the solvent and polymer, respectively, μ_i^j is the infinite-dilution chemical potential of component i in pure liquid phase j , and V^j is the molar volume of liquid phase j . A positive partition coefficient ($\log P > 0$) indicates that at equilibrium the contaminant is more likely to partition to the solvent phase, while a negative value ($\log P < 0$) indicates that the contaminant prefers to remain in the polymer matrix. In the context of contaminant removal from plastics, we thus hypothesize that calculating $\log P$ can predict the efficiency of contaminant removal to determine the selection of appropriate solvents to remove one or more targeted contaminants from a given polymer. We note that this approach is based on equilibrium thermodynamics, without consideration of kinetic factors that may substantially impact low-temperature leaching.⁴³ Consequently, $\log P$ values determine the maximum theoretical removal capacity to guide solvent selection, whereas practical contaminant removal rates could be governed by mass transfer kinetics that are not considered here.

Thermodynamic property prediction using COSMO-RS

The conductor-like screening model for real solvents (COSMO-RS) is used to predict thermodynamic properties that are relevant to contaminant removal, including polymer solubilities, solvent miscibility with contaminants, and contaminant $\log P$ values.⁴⁴ COSMO-RS is well-suited for this study because it can predict thermodynamic behavior while requiring only input density functional theory calculations (DFT), without requiring new experimental data, which is advantageous for considering the properties of chemically diverse contaminants. Several studies have shown COSMO-RS to predict organic compound solubility and partitioning in liquid phases in strong agreement with experimental measurements, with performance comparable to or more accurate than alternative equation-of-state or continuum-solvation methods.^{45–49} Finally, this approach enables the use of our previous methodology^{38,39} to represent the physicochemical properties of polymers using DFT calculations of representative oligomers, as detailed in the SI, to predict contaminant removal.

The COSMO-RS computational property predictions involve several steps. Fig. 2 depicts the methodological workflow adopted to compute the necessary molecular properties of each contaminant, polymer, and solvent employed in this study. First, each selected molecular structure is input to DFT calculations to obtain a screening charge density profile.

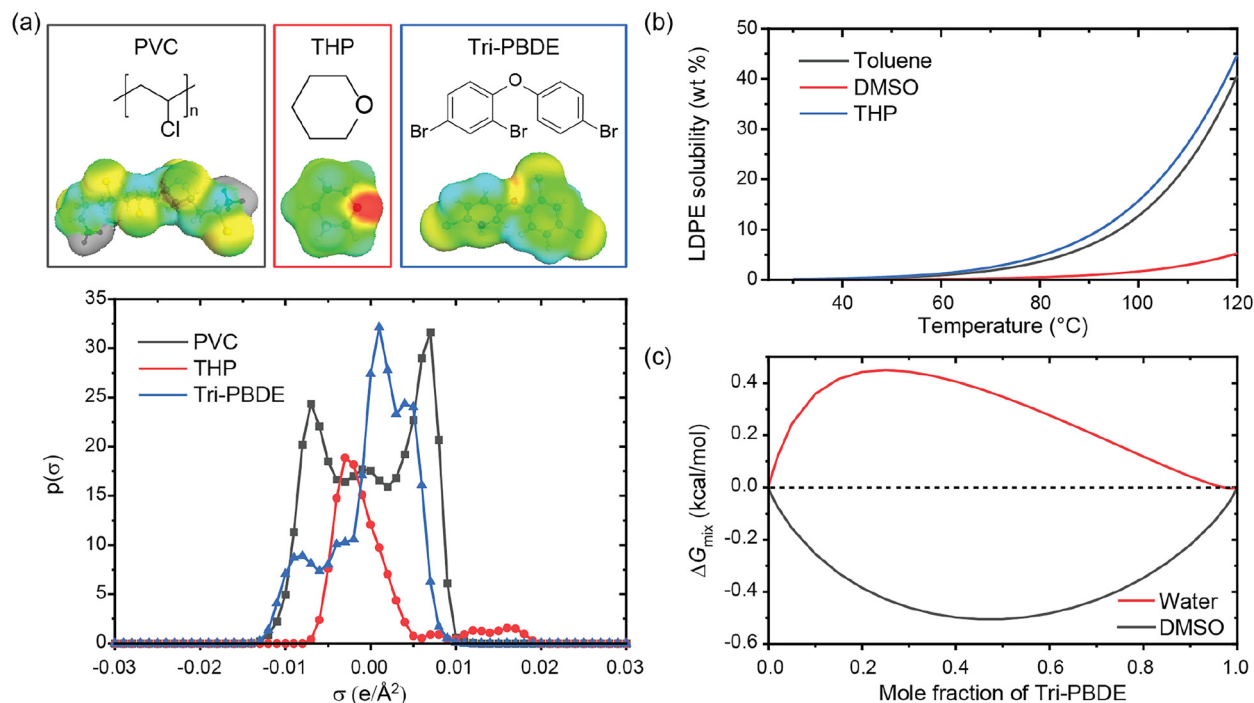


Fig. 2 Overview of COSMO-RS calculations. (a) Screening charge densities of three representative species – a polymer (PVC), solvent (THP), and contaminant (Tri-PBDE) – and corresponding σ -profiles, which are used as input to all other calculations. (b) Temperature-dependent polymer solubilities shown for low-density polyethylene (LDPE) in three solvents as an example. (c) Liquid–liquid equilibrium calculation to determine contaminant miscibility. The example shows the free energy of mixing for Tri-PBDE in either dimethyl sulfoxide (DMSO; fully miscible) or water (immiscible).

Examples of the molecular structures are shown in Fig. 2a for an example polymer (PVC), solvent (tetrahydropyran, THP), and contaminant (2,4,4'-tribromodiphenyl ether, Tri-PDBE). This step includes a geometry optimization in implicit water using the conductor-like polarizable continuum model and a single-point calculation in the infinite dielectric constant limit, and is performed with Gaussian 16 at the BVP86/TZVP/DGA1 level of theory.⁵⁰ Examples of the resulting screening charge densities and corresponding σ -profiles for the polymers, solvents, and contaminants from this step are also depicted in Fig. 2a. Screening charge densities for polymers and solvents were obtained from our prior work,^{38,39} whereas screening charge densities for contaminants were newly computed using a single energy-minimizing conformer per contaminant molecule. A summary of our previously developed method for obtaining screening charge densities for polymers^{38,39} is presented in the “polymer and solvent modeling” section of the SI, and a complete list of the 32 solvents considered in this work is included as part of Table S1. Second, COSMO-RS statistical thermodynamics calculations use the screening charge density to calculate the chemical potentials of the compounds. Polymer solubilities, solvent miscibility, and contaminant partition coefficients are then calculated *via* solid–liquid and liquid–liquid equilibrium calculations.^{51,52} In this study, a contaminant is considered miscible in a solvent if the predicted liquid–liquid solubility at

a given temperature exceeds 15%. Given that most contaminants exhibited high solubility values (approaching 100%) across multiple solvents, miscibility was treated as a binary property: “yes” for solubility values greater than 15% and “no” otherwise, with this threshold chosen as a conservative value to account for any potential simulation error. All COSMO-RS calculations were performed using the COSMOtherm19 software with the BP_TZVP_19 parameterization,⁵³ although the BP_TZVP_FINE_19 parameterization was used when predicting partition coefficients in solvent mixtures for better accuracy.⁴⁹ Fig. 2b shows an example of the temperature-dependent solubility curve of polyethylene in three different solvents obtained from COSMO-RS. Similarly, Fig. 2c shows an example of the free energy of mixing of an example contaminant (tri-PDBE) in two solvents at room temperature to illustrate characteristic curves for a solvent in which the contaminant is fully miscible (DMSO) and immiscible (water). We similarly compute $\log P$ values using COSMO-RS,^{54,55} although these are not shown in Fig. 2.

Experimental contaminant removal

Experiments were carried out for the phthalate removal case study. Benzyl-*n*-butyl phthalate (BBP) was selected for it due to its characteristic peak appearance in gas chromatograms. It appears as one clear peak while some other phthalates have multiple peaks. BBP was physically blended with polyvinyl

chloride (PVC, Sigma Aldrich average $M_w \sim 80\,000$ and average $M_n \sim 47\,000$) at a ratio of 1 : 2.5 by mass, respectively, to a total mass of 100 g, then melt-compounded in a twin-screw extruder at 150 °C and a screw speed of 30 rpm under mild conditions to minimize thermal degradation. The produced cylindrical filament was 2–2.2 mm in diameter which was cut into 4.8–6 mm long pieces. Three consecutive solvent washes were performed in each solvent at each temperature. Xylenes (Aqua Solutions, Reagent ACS Grade) and dodecane (ThermoFisher Scientific, mixture of isomers) were selected as solvents to assess BBP removal effectiveness at room temperature and at 130 °C. The starting polymer mixture mass was 1.0 g and solvent mass was 75 g. Experiments were conducted in glass pressure tubes to avoid solvent evaporation. The solution and polymer were allowed to stir for 1 hour per wash. After each wash, the polymer was recovered, dried for three hours at 100 °C in a vacuum oven and transferred to the next wash with 75 g of new solvent. For experiments at 130 °C, glass tubes were quenched in ice for 5 minutes prior to solvent recovery. Prior to analysis, all samples were filtered using 0.22 μm filters to ensure no PVC was retained in the injected samples. Recovered solvents were analyzed through gas chromatography (Shimadzu GC-2010) with a Phenomenex ZB-5HT Inferno™ column and N_2 as the carrier gas. A calibration curve was used for determining the concentration in samples. The concentrations were converted into mass and compared to phthalate mass in the added polymer mixture to calculate the percent recovery. For experiments at 130 °C, some PVC dissolved (as expected; *vide infra*) and precipitated during quenching. The solution was filtered using a 100–200 μm wire mesh to separate the solution and the PVC. Precipitated PVC was collected from the mesh while any remaining dissolved PVC and BBP was retained in solution. The collected PVC was then dried and transferred to the next wash.

Results and discussion

Solvent selection criteria for contaminant removal

For a given set of contaminants, we propose different solvent selection criteria based on the characteristics of the two contaminant removal processes in Fig. 1. These selection criteria depend upon contaminant miscibility, the contaminant partition coefficient, and polymer solubilities in the possible solvents.

The selection criteria for leaching are:

1. The target contaminants must be miscible in the selected solvent (*i.e.*, the solvent can dissolve the contaminants).
2. Contaminants should have positive partition coefficients ($\log P > 0$); higher values are preferred to limit retention within the polymer.
3. Ideal solvent candidates will swell but not dissolve the polymer.

These criteria are designed to ensure the preferential retention of the contaminants within the chosen solvent without dissolving the polymer itself. Swelling of the polymer is

desired to maximize solvent contact with contaminants within the polymer matrix and to enable efficient mass transport into the solvent phase.

The selection criteria for the STRAP contaminant removal process are:

1. The solvent can dissolve both the polymer and target contaminants.
2. The solvent does not dissolve non-target compounds, such as other polymers in the plastic waste.
3. The polymer can be precipitated from solution either by decreasing the temperature (or by adding an antisolvent).
4. Contaminants should have positive partition coefficients ($\log P > 0$); higher values are preferred.
5. The contaminants do not precipitate (*i.e.*, they remain miscible in the solvent) under the conditions in (4) that induce polymer precipitation.

These criteria are designed to ensure that target contaminants dissolve along with the target polymer and then preferentially remain in the solvent phase when the polymer precipitates. In this work, we assume that precipitation is induced by a decrease in temperature, as opposed to adding an antisolvent, due to more favorable environmental and economic impacts associated with temperature-dependent precipitation.⁵⁶ Following our past work, we set a threshold of 1 wt% as the maximum computationally predicted solubility to achieve precipitation, although in practice lower solubilities are observed experimentally.^{31,32,38–40} Dissolved contaminants removed during this process can be potentially removed from the solvent *via* additional methods not explicitly discussed here, such as by adsorption.³³

It is computationally expensive to perform property predictions for every exact chemical structure because contaminants in plastic waste (including both BIAS and additives intentionally included in plastic products) are often mixtures of similar chemical compounds. Most of these chemical families include a wide range of different substances.¹⁰ To conduct feasible property calculations suitable for satisfying the above criteria, we first need to select a set of representative chemicals, then use their molecular structures to perform predictions of interest. In this work, we propose multiple case studies in which we select representative chemical structures for different classes of contaminants based on common compounds reported in literature. Specifically, given the large number of contaminants classified within each “priority group of chemicals” identified by Monclús and co-workers due to their significant risks to human health, safety, and the environment,¹⁰ a representative subset was selected for detailed investigation. This subset comprised of four BFRs, eight phthalates, and twenty-six PFAS. The rationale for selecting these compounds is provided within the corresponding case studies presented in the following sections. Their structures are then input to COSMO-RS to predict miscibility, solubility, and partition coefficients as detailed in the Methods to identify solvents suitable for contaminant removal *via* leaching or STRAP. Fig. 3 shows the chemical structures of polymers, example common solvents, and example contaminants studied below. The full list and

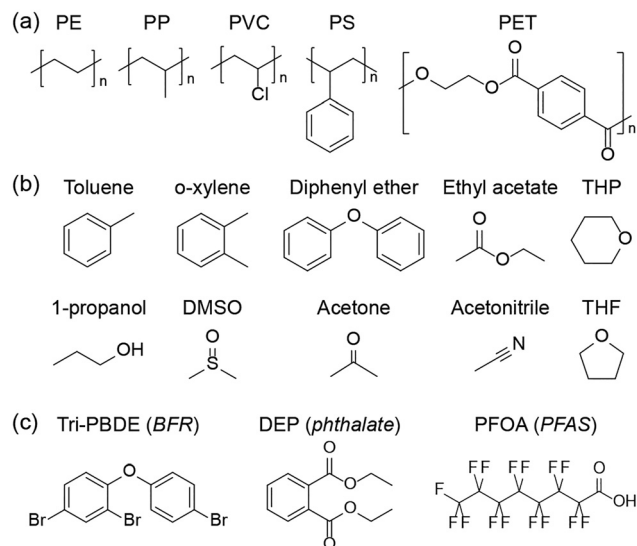


Fig. 3 Chemical structures of (a) polymers, (b) example common solvents, and (c) example contaminants considered in this work. PE: polyethylene, PP: polypropylene, PVC: polyvinyl chloride, PS: polystyrene, PET: polyethylene terephthalate, THP: tetrahydropyran, DMSO: dimethyl sulfoxide, THF: tetrahydrofuran, Tri-PBDE: 2,4,4'-tribromodiphenyl ether, DEP: diethyl phthalates, PFOA: perfluorooctanoic acid.

chemical structures of contaminants studied in the work are included in the SI.

Case study 1: BFR leaching from plastic products. As a first case study, we consider solvent selection for the leaching of brominated flame retardants (BFRs) from common polymers including polystyrene (PS), polypropylene (PP), polyethylene (PE), and polyethylene terephthalate (PET). We compare computational predictions to known literature data and identify solvent candidates for BFR removal. BFRs are synthetic chemicals widely used as additives in plastic materials to reduce product flammability.^{13,57} Furthermore, BFRs are incorporated as flame-retardant additives into consumer products, particularly electrical and electronic equipment, many of which are predominantly composed of PS.¹⁰ However, they are persistent organic pollutants with bio-accumulative properties.⁵⁸ They have been reported at hazardous concentrations in aquatic environments and soils where they can accumulate on PP, PE, and PET microplastics which constitute a significant fraction of microplastics detected in terrestrial and aquatic environment.⁵⁹ Exposure to BFRs has been associated with serious health concerns, including cancer, neurobehavioral disorders, and reproductive issues.⁵⁷ Common end-of-life plastic management processes, such as mechanical recycling or incineration, are not able to separate BFRs from the target polymer resins, thus posing significant environmental and health risks by releasing hazardous halogenated compounds.⁶⁰ Conversely, recent work has indicated the feasibility of solvent-based methods for removing BFRs from waste plastics obtained from electrical and electronic equipment.⁶¹

Fig. 4 shows selected common BFRs and their chemical structures, along with COSMO-RS σ -profiles to illustrate the

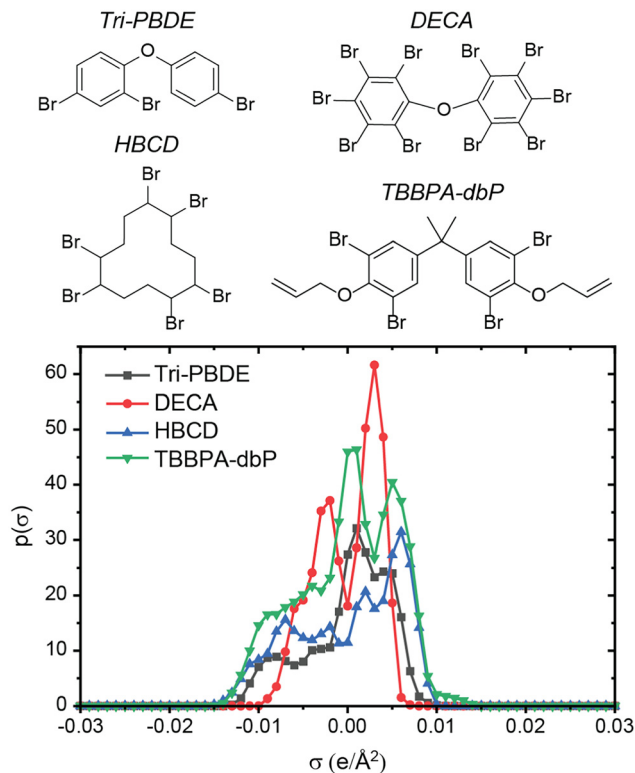


Fig. 4 Chemical structures of four BFRs modeled and used for qualitative validation of leaching approach. Corresponding COSMO-RS σ -profiles for the four BFRs are shown at bottom. The variations in the σ -profiles illustrates the chemical diversity within this class of contaminants.

distinct chemical properties for these molecules. These BFRs were selected due to the availability of literature experimental data reporting the solvent extraction of BFRs from common polymers.^{62,63} These literature references provide the opportunity to perform computational screening to select solvents according to the leaching process in Fig. 1, then compare computational predictions to literature results. Hexabromocyclododecane (HBCD), decabromodiphenyl ether (DECA), and 2,4,4'-tribromodiphenyl ether (Tri-PBDE) are designated as “chemicals of concern” in the PlastChem database, reflecting their persistence, bioaccumulation potential, carcinogenicity, mutagenicity, and reproductive toxicity.¹⁰ In addition, polybrominated diphenyl ethers (PBDEs) form a major class of BFRs and have been detected at extreme concentration levels in waste plastics, particularly ocean plastics.⁶³ Based on the BFRs and polymers reported in the literature, we thus first performed computational predictions of $\log P$ values for BFRs, polymer solubilities, and BFR miscibility to qualitatively compare with experimental observations in the literature.

Table 1 presents data on the experimental solvent extraction of the four BFRs introduced in Fig. 4 from different polymers^{62,63} and compared to our computational predictions. HBCD, DECA, and TBBPA-bis-2,3-dibromopropylether (TBBPA-dbp) were extracted from PS obtained from discarded televi-

Table 1 Comparison of $\log P$ calculations with experiment observations of BFR removal from several different polymers reported in the literature.^{62,63} All but the extraction of HBCD from PS polymer using toluene/1-propanol solvent mixture at 60 °C show agreement between computational predictions and experimental results

BFR	Polymer	Temperature (T)	Literature experiments	Computational prediction		
			Selected solvent (extraction efficiency)	Polymer solubility at T	BFR miscibility at T	Predicted $\log P$
Tri-PBDE	PET	RT (25 °C)	Acetonitrile (78.0%)	0.0%	Yes	0.4
	PE		Toluene (111.4%)	0.1%	Yes	1.5
	PS		Ethyl acetate (71.4%)	5.5%	Yes	1.2
	PP		Methanol (2.1%)	0.0%	Yes	−0.1
HBCD	PS	60 °C	Toluene (138.4%)	0.8%	Yes	1.7
DECA			Toluene + 1-propanol (1 : 1) (7.7%)	3.1%	Yes	1.0
TBBPA-dbp			Toluene + 1-propanol (1 : 1) (100%)		Yes	0.6
			Acetonitrile (93.8%)	0.1%	Yes	2.3

sion and personal computer housings at 60 °C using different solvents while tri-PBDE that first adsorbed to PET, PE, PS, and PP polymers from aqueous solution (to represent contaminated environmental plastics) was extracted at room temperature. Experimental procedures used by Singla *et al.*⁶³ and Altwaiq *et al.*⁶² for these samples are summarized in the SI. In both experimental studies, the extraction efficiency for each contaminant was calculated as the percentage of the contaminant mass recovered in the organic extract from the plastic relative to the total amount of contaminant introduced onto the plastics. This was quantified using GC/MS or HPLC/UV. Based on our framework for extracting contaminants *via* leaching, we expect that a target BFR will be extracted if $\log P > 0$ in the selected solvent, the target BFR is miscible in the selected solvent, and the solubility of the polymer in the selected solvent is low, thus fitting the criteria for process 1 in Fig. 1.

All experimental results from the extraction of Tri-PBDE using the solvents listed in Table 1 are consistent with our computational predictions. For example, the reported high extraction efficiency of Tri-PBDE with toluene (138.4%) aligns with our predicted values, which indicated favorable miscibility, low polymer solubility (0.8 wt%), and a $\log P$ of 1.7. Conversely, the low experimental extraction efficiency observed with methanol (2.1%) agrees with our computational prediction that methanol is a poor solvent for this process, as evidenced by the negative partition coefficient ($\log P < 0$), despite satisfying the other two criteria of BFR miscibility and low polymer solubility. Based on the results summarized in Table 1, extraction efficiencies exceeding 100% are likely attributable to experimental uncertainties in mass balance calculations or may indicate that the extraction solvents co-extracted unidentified contaminants or compounds originating from untargeted sources, such as plastic materials used during the experimental procedures. In general, higher $\log P$ values correlate with higher extraction efficiencies. Similarly, Table 1 shows high experimental extraction efficiencies for DECA (100%) and TBBPA-dbp (93.8%) that are also in good agreement with our predictions, supported by $\log P$ values of 0.6 and 2.3, respectively, along with compliance with low polymer solubility and BFR miscibility requirements at 60 °C. Only the extraction of HBCD from polystyrene using a toluene/

1-propanol mixture deviated from computational expectations; while our predictions suggested preferential extraction at 60 °C, the experimental efficiency was low (7.7%). This discrepancy may arise from kinetic effects since partition coefficients are computed assuming equilibrium, and it is possible that extending the extraction time beyond the 2 hours adopted by Altwaiq *et al.* could improve the extraction efficiencies as shown for phthalate extraction in Case study 2 (described below). Overall, the data suggest that computational $\log P$ predictions correlate well with experimental findings, highlighting the potential of our computational approach to guide solvent selection for BFR removal.

Given the strong agreement between literature data and our computational predictions, we subsequently screened 32 common solvents to identify those capable of simultaneously removing all four selected BFRs from the polymers listed in Table 1 (PS, PET, PE, and PP) at room temperature. The objective was to identify solvents that satisfy the established screening criteria while enabling the removal of all four BFRs rather than optimizing for a single compound. The predicted $\log P$ values, BFR–solvent miscibility, and polymer solubilities across different solvents are summarized in Tables S1–S4. The results reveal that, even if several solvents have the potential to extract all four BFRs, distinct trade-offs are evident. Specifically, solvents demonstrating high efficacy for one BFR often perform sub-optimally for others. Representative examples of these trade-offs for BFR removal from polystyrene using five selected solvents are illustrated in Fig. 5. While all five solvents meet the BFR miscibility criteria, variations arise in their ability to satisfy the $\log P$ requirements across all BFRs. For instance, *N,N*-dimethylformamide (DMF) and dimethyl sulfoxide (DMSO) exhibit favorable partitioning for HBCD and TBBPA-dbp but are less effective for DECA, as reflected by its low $\log P$ value. Conversely, tetrahydropyran (THP) demonstrates favorable partitioning for Tri-PBDE, HBCD, and TBBPA-dbp, but not for DECA, which displays a negative $\log P$ value. This example highlights that selecting solvents for BFR removal by maximizing the partitioning of a single compound may not be optimal for removing all possible BFRs. In contrast, even they do not lead to the highest $\log P$ values for any individual BFR, benzene and diphenyl ether

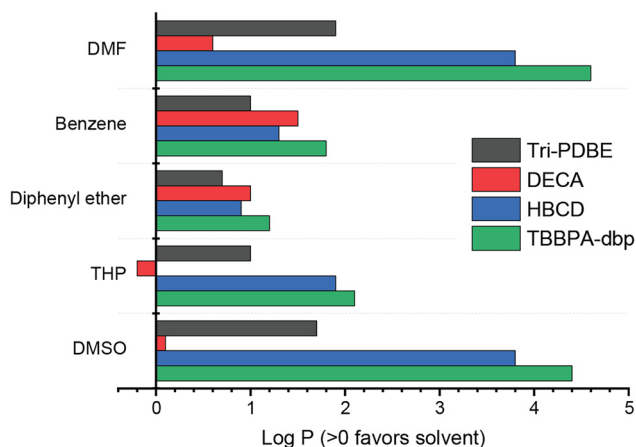


Fig. 5 Observed trade-offs during solvent selection for the removal of all four BFRs from polystyrene. $\log P < 0$ indicates poor solvents for specific BFR removal (partitioning into the polymer phase) while $\log P \geq 0$ indicates good solvent for specific BFR removal (partitioning into the solvent phase).

exhibit consistently favorable partitioning across all four BFRs, suggesting that these solvents are preferred for more broadly removing a wider range of BFRs; selection between these solvents may ultimately depend on economic, availability, environmental, and safety considerations.⁴⁰

Case study 2: phthalate removal from PVC. The first case study established that computational predictions align qualitatively with experimental literature results for the leaching of BFRs from varied polymers at temperatures close to, or somewhat above, room temperature. In the second case study, we extended the computational screening protocol to identify suitable solvents for removing phthalates from PVC resins. Phthalates are a family of synthetic chemical compounds used in plastics, solvents, and personal care products. In plastic products, phthalates are widely used as plasticizers in the production of flexible and durable plastic materials, particularly PVC.^{64,65} Phthalates are often physically integrated into plastics rather than chemically bound, which means they can migrate out of products through wear, heating, or contact with liquids.⁶⁶ Short-chain phthalates, such as dimethyl phthalate (DMP), are more volatile and can evaporate into the air, whereas longer-chain phthalates, like di(2-ethylhexyl) phthalate (DEHP), tend to leach from plastics into surrounding environments over time.^{64,67} Phthalates are also endocrine-disrupting substances associated with numerous adverse health consequences, including elevated risks of cancer, and consequently have been subjected to increased regulation for recycled plastics, motivating their removal.^{68,69} Since there are hundreds of different phthalate compounds, selecting a representative set of structures is essential for practical analysis and computational modeling. Table S5 presents the eight selected phthalates for this study and their common applications in commercial products.^{67,69} These representative phthalates were chosen based on their prevalence in commercial applications, structural diversity, and relevance to environmental

and health concerns.⁶⁷ By focusing on a subset of commonly used phthalates, we can effectively predict their properties and interactions with plastic resins and in the meantime maintain the generalizability of the findings. The chemical structures and σ -profiles showing the chemical diversity of the eight representative phthalates are presented in Fig. S1 and S2 of the SI.

In this second case study, we sought to perform both computational predictions and experiments to evaluate both distinct processes (leaching and STRAP contaminant removal) presented in Fig. 1. The primary difference between these processes is that STRAP contaminant removal involves a high-temperature polymer dissolution step followed by low-temperature polymer precipitation, whereas leaching is performed at a single temperature and does not involve polymer dissolution or precipitation. We note that STRAP also requires considering only the selective dissolution of a target polymer when multiple polymer components are present, although that consideration is omitted for this case study (but is included as part of Case study 4, *vide infra*). To identify suitable solvents that facilitate phthalate removal *via* both processes, we used COSMO-RS to compute $\log P$ values (relative to PVC), phthalate miscibility in each solvent, and PVC solubility at multiple temperatures. A library of 32 common solvents was considered. Table 2 presents computational prediction results for a subset of solvent candidates; additional $\log P$ data are presented in Table S6. We placed an upper bound of 130 °C on the screening temperature for polymer solubility because higher temperatures may lead to degradation of some plastics.

We sought to identify a solvent capable of performing both leaching at room temperature and STRAP contaminant removal at a high temperature to compare contaminant removal efficiencies with both processes. According to our solvent selection criteria for both leaching and STRAP, the ideal solvent should be miscible with all eight phthalate compounds considered and should promote positive $\log P$ values (with higher values preferred) at room temperature. Additionally, we also require a low PVC solubility at room temperature in the solvent, with a high solubility at elevated temperatures to enable polymer dissolution as part of STRAP. Based on past studies,^{31,32,38–40} we defined the low temperature solubility threshold as <1 wt% and the high temperature solubility threshold as >15 wt%. From the COSMO-RS calculations, nine solvents meet the $\log P > 0$ and miscibility requirements: chloroform, tetrahydrofuran, tetrahydropyran, 2,3-dihydropyran, acetone, toluene, ethyl acetate, *o*-xylene, and diphenyl ether. These solvents (the first nine listed in Table 2) are thus expected to be suitable for leaching. In addition to having positive mean $\log P$ values, these solvents have positive $\log P$ values for each individual phthalate, suggesting that these solvents are capable of extracting all eight representative phthalates. This result is consistent with the structural resemblance among the eight phthalates, as illustrated by their molecular structures and σ -profiles in Fig. S1 and S2. However, not all solvents with positive mean $\log P$ values have positive $\log P$ values for all phthalates; DMSO and triethylamine are

Table 2 COSMO-RS results for solvent screening for phthalate removal from PVC. The elevated temperature T_h is defined to be 1 °C lower than the boiling point (BP) of the solvent, with an upper bound of 130 °C. All log P calculations were performed at room temperature. *ortho*-Xylene and dodecane (bold) are used in validation experiments

Lab name	BP (°C)	T_h (°C)	All log $P > 0$	Mean log P	Phthalate miscibility		PVC solubility (wt%)	
					RT	T_h	RT	T_h
Chloroform	61.1	60.1	Yes	1.94	Yes	Yes	1.1	4.2
Tetrahydrofuran	65	64.0	Yes	1.21	Yes	Yes	8.9	19.1
Tetrahydropyran	88	87.0	Yes	0.99	Yes	Yes	3.9	17.9
2,3-Dihydropyran	86	85.0	Yes	0.88	Yes	Yes	1.9	13.6
Acetone	56	55.0	Yes	0.81	Yes	Yes	10.9	19.8
Ethyl acetate	77.1	76.1	Yes	0.71	Yes	Yes	4.5	16.3
<i>o</i>-Xylene	144.5	130.0	Yes	0.61	Yes	Yes	0.5	19.8
Toluene	110.6	109.6	Yes	0.72	Yes	Yes	0.7	14.8
Diphenyl ether	258	130.0	Yes	0.29	Yes	Yes	0.3	11.5
Triethylamine	89	88.0	No	0.38	Yes	Yes	0.5	4.7
Dimethyl sulfoxide	189	130.0	No	0.23	No	Yes	8.4	37.2
Hexane	68.7	67.7	No	−0.04	Yes	Yes	0.1	0.6
<i>tert</i> -Butanol	82.4	81.4	No	−0.04	Yes	Yes	0.2	3.3
Ethanol	78.2	77.2	No	−0.32	Yes	Yes	0.2	2.8
Dodecane	216.3	130.0	No	−0.45	No	Yes	0.0	2.8
Acetic acid	117.9	116.9	No	−0.77	No	Yes	0.1	5.2
Ethylene glycol	197.3	130.0	No	−2.12	No	No	0.0	1.8
Water	100	99.0	No	−8.02	No	No	0.0	0.0

notable counterexamples, indicating the need to evaluate log P for all possible contaminants, not just individual contaminants, to ensure a high probability of removal.

Among the evaluated solvents, only one solvent – *ortho*-xylene (*o*-xylene) – satisfies the PVC solubility requirements at both high and low temperatures. Toluene meets the low-temperature solubility threshold and is also very close to meeting the high-temperature solubility threshold with a predicted PVC solubility of 14.8 wt%, while diphenyl ether also meets the low-temperature solubility threshold and has a moderate PVC solubility of 11.5 wt% at a high temperature. These solvents are thus likely to be suitable for both leaching and STRAP contaminant removal. Of the solvents considered, *o*-xylene was chosen as a suitable solvent for experimental evaluation of both STRAP contaminant removal and leaching, since it exhibits a favorable mean log P value of 0.61 at room temperature, has the highest predicted PVC solubility of the three potential STRAP solvents at the maximum temperature of 130 °C, and has minimal predicted PVC solubility at room temperature.

We conducted two sets of experiments on a sample of PVC and phthalate to verify the ability of *o*-xylene to remove phthalates from PVC resins. The sample is a physical mixture of PVC powder and liquid butyl benzyl phthalate (BBP) with a mass ratio of 2.5 : 1. This ratio was determined based on the typical concentration of phthalate additives in industrial PVC materials.⁷⁰ The mixture was extruded as filaments and cut into short pieces (Fig. 6a). Experiments used a mixture of xylene isomers (xylenes) due to their availability and the expectation that they would behave similarly to *o*-xylene alone for the proposed contaminant removal processes. The first set of experiments used xylenes to remove BBP from the PVC sample based on the criteria described above. The second set of experiments used dodecane as a solvent to compare to xylenes

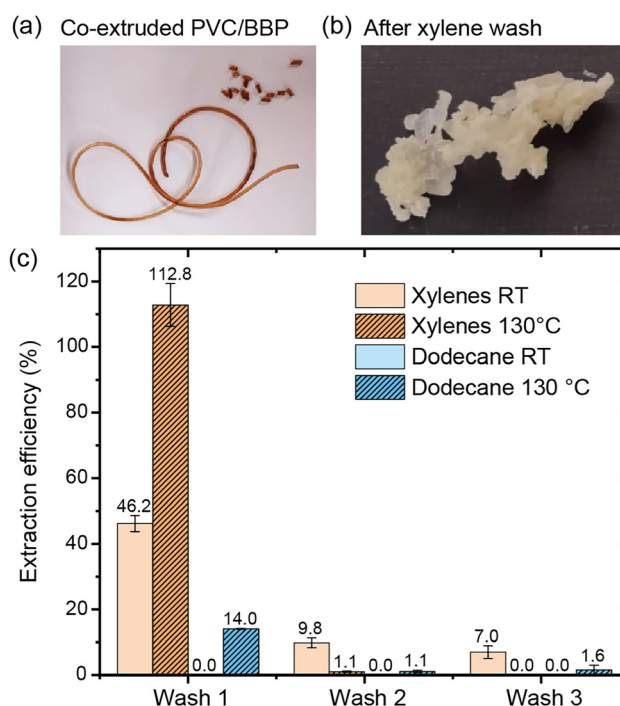


Fig. 6 Experimental phthalate removal using BBP as a representative phthalate. (a) Image of the extruded PVC and BBP sample as a filament, which was cut into smaller pieces which are also shown. (b) PVC after BBP removal via xylene leaching; the sample changed color. (c) BBP extraction efficiency at room temperature and 130 °C using a solvent that is predicted to promote contaminant removal (xylenes) and a control solvent that is not predicted to promote contaminant removal (dodecane) based on the data in Table 2. Experiments at room temperature model the leaching process because PVC does not dissolve in either solvent. Experiments at 130 °C model STRAP contaminant removal because PVC dissolves in xylenes at this temperature. Error bars indicate the standard deviation across three replicate experiments.

because Table 2 indicates that dodecane should not promote phthalate removal based on the negative $\log P$ values. We employed dodecane as a negative control solvent to test our computational predictions. Three experimental wash steps were adopted. The first wash step (wash step 1) involved immersing the sample in fresh solvent for 1 hour, after which further analysis quantified the fraction of BBP that was removed. Next, the sample after wash step 1 was dried and then immersed in a fresh solvent for another 1 hour after which the fraction of removed BBP was quantified. A similar approach was adopted for the wash step 3. Both sequences of wash steps were performed in xylenes and dodecanes at two temperatures to model the two contaminant removal processes. Experiments were performed at room temperature to model the leaching process because PVC dissolution was not expected at this temperature in either solvent. Experiments were also performed at a high temperature of 130 °C to model STRAP contaminant removal because PVC dissolution (in *o*-xylene) was expected at this temperature. All experiments were done in triplicate.

Fig. 6 shows the experimental measurements of phthalate mass recovered from the PVC removal experiments at room temperature and at 130 °C as well as an image of the sample after BBP removal. At room temperature, an average of $63.0 \pm 5.2\%$ (with the average and standard deviation computed across three replicate experiments) of the phthalates was removed in the 3-hour experiment (*i.e.*, after the three wash steps), with decreasing percentages recovered after each wash step. We note that the predicted $\log P$ value for *o*-xylene is modest (with a mean value < 1 and value for BBP of 0.28; see Table S6). As shown in Table 1, similar predicted $\log P$ values led to modest extraction efficiency of BFRs at room temperature (*e.g.*, acetonitrile had a $\log P$ value of 0.4 and led to 71.8% extraction efficiency for leaching tri-PBDE from PET). Fig. 6c shows that performing the high-temperature STRAP experiments significantly improved BBP extraction efficiency to an average of $113.9 \pm 6.4\%$, with nearly all of the contaminant removed in the first wash step. In contrast, dodecane was ineffective for phthalate removal as expected. At room temperature, dodecane was completely unable to extract BBP from PVC in agreement with our computational predictions, while at 130 °C the average extraction efficiency of BBP from PVC was $16.7 \pm 1.6\%$, indicating a substantially lower extraction efficiency compared to *o*-xylene. Table 2 also indicates that at the elevated temperature BBP is miscible in dodecane, yet was still not readily removed, highlighting the utility of $\log P$ values for predicting contaminant removal. The slightly higher fraction of BBP that was removed at the high temperature is likely due to the slight dissolution of the polymer, but the low removal indicates that dissolved BBP preferentially precipitates with the PVC in dodecane.

Overall, the increased extraction efficiency at 130 °C suggests that polymer dissolution as part of STRAP generally leads to higher contaminant removal than leaching. These results validate the ability of xylenes to remove phthalates from PVC resins, and more broadly serve as proof-of-concept

for the computational methodology. However, Fig. 6c also shows continuous removal of BBP after each 1 hour solvent wash at room temperature, whereas the predicted room temperature $\log P$ value of 0.6 should indicate an $\sim 4\times$ preference of BBP for the solvent *vs.* the polymer. Given the large amount of solvent compared to the amount of polymer sample (75 g solvent, 1 g polymer), these experimental observations indicate a lower extraction efficiency than would be expected based on the predicted $\log P$ value, which is likely due to mass transfer kinetics governing contaminant migration from the polymer phase into the solvent phase.⁴³ Such kinetic effects would be expected to be prominent at lower temperatures and in the absence of polymer dissolution and will depend on process variables such as contact time, temperature, and concentration gradients, which are not captured within the current equilibrium-based modeling approach. Therefore, we expect that operating conditions such as time and temperature could be further investigated to improve the efficiency of the process for actual deployment in more realistic scenarios.

Case study 3: STRAP with PFAS removal from LDPE. Case studies 1 and 2 established that predicted $\log P$ values can be used to guide solvent selection for contaminant removal *via* leaching processes, with solvents exhibiting high $\log P$ values removing contaminants in experiments as expected. These promising results motivate additional case studies that integrate contaminant removal with STRAP to enable the simultaneous recovery and purification of target polymers during a recycling process. As an additional example of integrating STRAP with contaminant removal for a larger category of contaminants, we use computational predictions to assist the design of a process for removing PFAS from plastic waste. PFAS are a diverse group of synthetic fluorinated compounds known for their remarkable stability, water and oil repellency, and resistance to heat and chemical degradation.⁷¹ These properties have made PFAS essential in applications such as non-stick cookware, waterproof fabrics, grease-resistant food packaging, and firefighting foams.⁷² However, their environmental persistence and bioaccumulation pose significant health risks, including links to cancer, liver toxicity, and immune system disruption.⁷² Common PFAS, including perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS), both containing eight carbon atoms, as well as shorter chain perfluoroalkyl acids (PFAAs) and perfluoroalkyl sulfonic acids (PFSAs), have been detected in aquatic systems, soils, and other environmental ecosystems.^{73,74} Although the use of PFAS as additives in plastic manufacturing has been increasingly regulated over the past two decades, their environmental ubiquity has led to their detection in commercial products.⁷⁵ This observation suggests that these compounds can migrate into plastic waste streams from surrounding environments through pathways such as aquatic systems, landfills, material recovery facilities, municipal waste collection systems, and open dumps. Recent studies have further confirmed the presence of PFAS in post-consumer recycled polyolefins used in food-contact applications.²¹ Consequently, PFAS must be removed during recycling processes to avoid their reintroduction into the plastic

packaging industry, particularly *via* post-consumer recycled resins.⁷⁶

Although thousands of PFAS have been identified by the US EPA,⁷⁷ their detection *via* experimental approaches have been limited. However, the FDA have been successful at detecting 26 PFAS concentration levels in food samples purchased at grocery stores in the US.^{78,79} Therefore, in this work, we apply our contaminant removal strategy to this set of PFAS. Table S7 and Fig. S3 and S4 of the SI show the PFAS compounds and chemical structures, and their σ -profiles, grouped based on their structures and primary functional groups. All 26 PFAS considered are small molecules representative of non-intentionally added substances that might be found in post-consumer waste; however, we note that fluorinated polymers such as polytetrafluoroethylene are also incorporated into plastic products as processing aids⁸⁰ as well as anti-drip additives, particularly in polycarbonates (PC) and PC/acrylonitrile butadiene styrene blends^{81,82} and may potentially be removed by similar solvents selected to remove the 26 target PFAS.

We considered the same library of 32 common solvents used in previous case studies for removing PFAS from low-density polyethylene (LDPE) as a reference polymer. As shown in Fig. S3 of the SI, some PFAS exist as ionized compounds.

Therefore, in the case of PFAS, distribution coefficients ($\log D$ values) which account for multiple ionization states are determined using COSMO-RS and used in place of their partition coefficient ($\log P$); these quantities have similar interpretations during solvent selection.⁸³ We performed COSMO-RS predictions of the distribution coefficients of all 26 PFAS in each solvent relative to LDPE, determined the miscibility of PFAS and each solvent, and computed LDPE solubility at multiple temperatures in the solvents. For STRAP contaminant removal, we require a high LDPE solubility at the elevated temperature and low LDPE solubility at room temperature to achieve temperature-based precipitation, with a mean $\log D$ value that is high at both temperatures to ensure preferential partitioning into the solvent of as much PFAS as possible.

We divided the 26 PFAS compounds into three different groups based on structural similarity (Fig. S3): PFAS within each group had similar chemical structures and consequently σ -profiles (Fig. S4), suggesting that they would be removed from LDPE resins using similar solvents. Fig. 7 presents the average σ -profiles and representative structures for each group, showing how the key functional groups in each PFAS group contribute to slight differences in σ -profiles. For example, group 2 is distinguished by the presence of sulfonic acid groups that contribute to a unique shoulder in the average σ -profile as a distinguishing feature of this group. Table 3 shows solvent screening and property calculation results for each group. THP is identified as the best solvent to remove PFAS in groups 1 and 2 as it has excellent mean $\log D$ values while leading to low LDPE solubility at room temperatures. Toluene is the best choice for group 3, since it satisfies positive $\log D$ requirements and has a strong temperature dependence of LDPE solubility (*i.e.*, able to dissolve LDPE at an elevated temperature and precipitate at room temperature), which is beneficial for STRAP.

In practice, the set of PFAS present in any given plastic waste may be unknown, so ideally single solvents capable of removing all PFAS would be identified. THP, toluene, and cyclohexanol were identified as effective solvents capable of removing all 26 PFAS by satisfying all the solvent screening criteria *via* the STRAP approach. THP exhibited consistently positive $\log D$ across all PFAS with an average of 5.1, satisfied the miscibility criteria for all PFAS, and showed low (0.1 wt%) and high (7.3 wt%) LDPE solubility at room and high temperatures, respectively. Similarly, toluene exhibited consistently positive $\log D$ across all PFAS with an average of 0.7, satisfied the miscibility criteria for all PFAS, and showed low (0.1 wt%) and high (22.6 wt%) LDPE solubility at room and high temperatures, respectively. Lastly, cyclohexanol exhibited consistently posi-

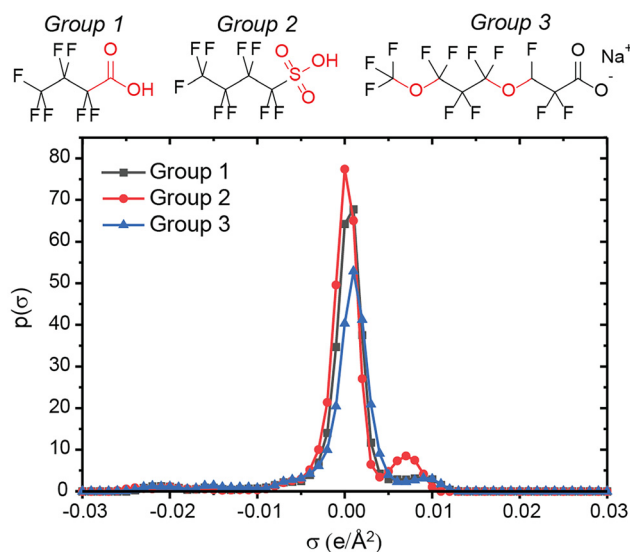


Fig. 7 Average σ -profiles and representative structures for each PFAS group. The distinguishing chemical feature for each group is highlighted in red. Group 1: carboxylic acid/carboxylate groups. Group 2: sulfonic acid/sulfonate groups. Group 3: perfluoroethers with carboxylic acid/carboxylate groups.

Table 3 Solvent screening results for the three groups of 26 total PFAS identified by FDA

Groups	Number of PFAS	Identified solvent	Mean $\log D$	LDPE solubility at RT	LDPE solubility at Th
1	12	THP	4.9	0.1 wt%	7.3 wt% (87 °C)
2	11	THP	5.1	0.1 wt%	7.3 wt% (87 °C)
3	3	Toluene	0.9	0.1 wt%	22.6 wt% (109.6 °C)

tive $\log D$ across all PFAS with an average of 3.8, satisfying the miscibility criteria for all PFAS, and showed low (0.0 wt%) and high (22.1 wt%) LDPE solubility at room and high temperatures, respectively. Owing to its high average $\log D$ value and pronounced temperature-dependent solubility behavior, cyclohexanol is identified as the preferred solvent for effectively removing all 26 PFAS from LDPE. These findings further support the potential for identifying a single solvent capable of extracting groups of structurally similar contaminants. The detailed predicted miscibility and $\log D$ values of all 26 PFAS considered in this work are presented in the SI. These results can further guide the experiment of PFAS removal from LDPE resins.

Case study 4: STRAP with BFR, phthalates, and PFAS removal from mixed plastic waste. Having studied the removal of three different classes of contaminants in different polymer systems, in this final case study we apply the computational framework to understand the potential removal of contaminants during the separation of multicomponent plastics *via* STRAP. Specifically, we analyze two previously developed, experimentally tested STRAP processes for multicomponent plastic waste recycling by Sánchez-Rivera and colleagues.^{31,32} The first process involved the separation of a 4-component multilayer plastic film containing LDPE, ethylene vinyl alcohol (EVOH), PET, and ethylene vinyl acetate (EVA). In the first step of this process, toluene was used to selectively dissolve PE from the remaining polymers at 110 °C. The dissolved PE was then cooled to 35 °C to aid its precipitation. The second process involved the separation of the same multilayer film composition with the addition of a polyurethane (PU)-based ink component. The first step of this process used dodecane instead of toluene to selectively dissolve PE at 95 °C while avoiding the dissolution of the PU-based ink, which was removed in a further processing step. The PE was then recovered by precipitation at 35 °C. Neither of these studies explicitly considered the potential for contaminants (other than the ink) to be dissolved and removed during STRAP. Therefore, in this case study, we aim to identify the contaminants that would have been removed from the first STRAP step in both processes (toluene and dodecane) based on the sets of contaminants (BFRs, phthalates, and PFAS) studied in the previous case studies.

Table 4 compares $\log P$ values (or $\log D$ values for PFAS) for the removal of contaminants from PE during the first STRAP

step using either toluene or dodecane, since the same polymer resin was removed first in both processes. Based on these values, and miscibility criteria, toluene is expected to remove all contaminants considered from PE during STRAP. This result suggests that during the recycling of the multilayer plastic *via* STRAP studied by Sánchez-Rivera and colleagues,³¹ all four BFRs, eight phthalates, and 26 FDA-identified PFAS would have been removed, producing a recycled LDPE film with little to no traces of the studied contaminants. On the contrary, dodecane is less efficient in removing any of the contaminant considered in this study, as also discussed in the negative control experiments conducted for phthalates in PVC in Case study 2. A summary of these findings, showing the flow of polymers and the extraction of these contaminants *via* the STRAP process using these two solvents based on an ideal, equilibrium process as predicted by our thermodynamic modeling approach is presented in Fig. 8. Since toluene was replaced by dodecane to avoid the co-dissolution of PU-based inks during the recycling of multilayer plastic film containing PU-inks,³² this suggests that dodecane can be first used to recover PU-based ink-free LDPE and subsequently toluene can be used to further remove all the contaminants listed in this study from the residual solids if desired.

In experimentally validated STRAP processes, the target polymer typically remains swollen with solvent after precipitation (*i.e.*, to form a polymer gel), which is not considered in the calculations above or Fig. 8. Retained solvent could impact the partitioning of contaminants between the solvent and (swollen) polymer by making these two phases more chemically similar, such that multiple wash steps would be required to remove this residual contaminant. Indeed, similar behavior was observed by Wagner and coworkers during the recycling of PVC waste containing phthalates *via* a multistep solvent-based process.⁸⁴ To assess the impact of retained solvent, we performed additional $\log P$ (or $\log D$ for PFAS) calculations to predict the room-temperature partitioning of contaminants between the solvent phase and a model swollen polymer phase. Similar to the data in Table 4, we chose two representative contaminants from each class (Tri-PBDE/HBCD as BFRs, BBP/DiNP as phthalates, and PFOA/PFOS as PFAS) and computed $\log P$ or $\log D$ in toluene and dodecane relative to a “swollen” PE phase. To represent the “swollen” PE, we note that prior STRAP studies have reported that the precipitated polymer gel retained between 70–80 wt% solvent after separ-

Table 4 Solvent screening summary for contaminant removal from LDPE during multilayer film recycling *via* STRAP. $\log P$ is used for BFR and phthalates while $\log D$ is used for PFAS. “Yes” indicates that all contaminants within each contaminant class exhibit positive $\log P$ or $\log D$ values and are all miscible in the specified solvent, whereas “no” indicates that at least one contaminant in each class has a negative $\log P$ or $\log D$ and is immiscible in the specified solvent

STRAP process	Solvent (Th)	Polymer solubility (wt%)		Log P or log D all > 0			Contaminant miscibility		
		Th	35 °C	BFRs	Phthalates	PFAS	BFRs	Phthalates	PFAS
1 ³¹	Toluene (110 °C)	22.6	0.1	Yes	Yes	Yes	Yes	Yes	Yes
2 ³²	Dodecane (95 °C)	8.8	0.1	No	No	No	No	Yes	No

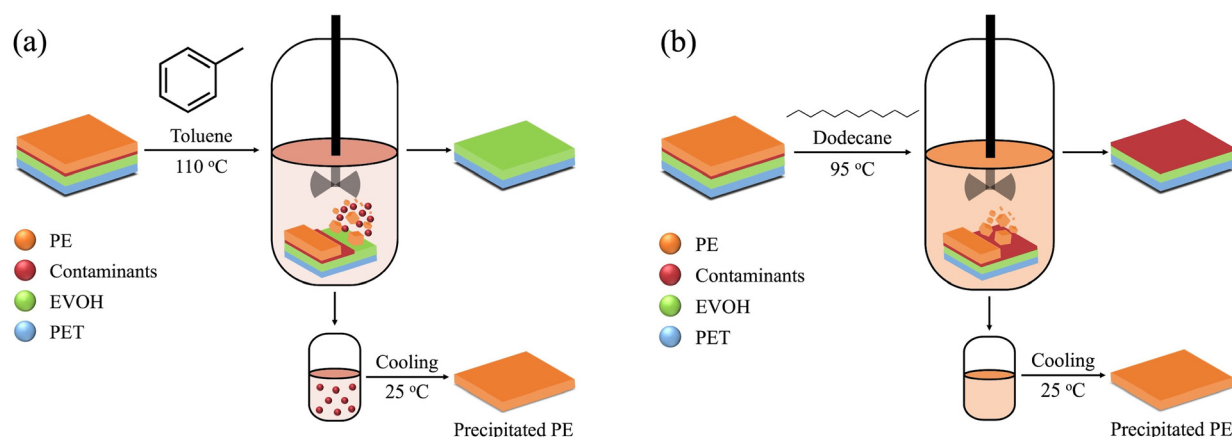


Fig. 8 Schematic showing the flow of polymers and contaminants during the recycling of multilayer plastic films via STRAP using (a) toluene, which favors the extraction of all contaminants considered in this work such that they end up with the solvent during PE removal, and (b) dodecane, which recovers contaminant-free PE.

Table 5 Partition ($\log P$ or $\log D$) coefficient prediction of select BFRs, phthalates, and PFAS between a pure solvent phase and either a pure polymer phase or a “swollen” polymer phase that is approximated as a mixture of the polymer and 70 wt% solvent at room temperature

Polymer phase	Solvent phase	Log P or log D					
		Tri-PBDE	DECA	BBP	DiNP	PFOA	PFOS
PE	Toluene	1.49	2.41	1.52	0.86	0.71	0.72
PE + toluene	Toluene	0.33	0.56	0.30	0.04	0.07	0.06
PE	Dodecane	−0.36	−0.68	−0.20	−0.20	−0.11	−0.13
PE + dodecane	Dodecane	−0.11	−0.20	−0.06	−0.06	−0.04	−0.04

PE – polyethylene, Tri-PBDE – 2,4,4′-tribromodiphenyl ether, DECA – decabromodiphenylether, BBP – butyl benzyl phthalate, DiNP – di-isononyl phthalate, PFOA – perfluorooctanoic acid, PFOS – perfluorooctanesulfonic acid.

ation of free solvent *via* vacuum filtration.³³ Therefore, we model the “swollen” PE gel phase as a mixture of PE with 70 wt% of the initial STRAP solvent, while the solvent phase is treated as pure.

Table 5 shows that, as expected, predicted $\log P$ (or $\log D$) values substantially change for all contaminants investigated. While the signs of all values remain the same, indicating no qualitative change in partitioning preference, all values are closer to zero when compared to values computed for partitioning from a pure polymer phase. These differences indicate that the presence of retained solvent would cause contaminants to distribute more evenly between the swollen PE and the solvent compared to the assumption of a dry polymer with minimal retained solvent. Consequently, in Fig. 8, retained solvent would lead to some contaminants remaining with the PE in the case of STRAP with toluene and some contaminants dissolving with the PE in the case of STRAP with dodecane. From a practical standpoint, this behavior implies that multiple washing steps following precipitation are likely required to obtain near contaminant-free PE after the STRAP process. Alternatively, additional processing steps could reduce the amount of retained solvent. For example, Yan *et al.* found that using a piston to mechanically compress the polymer during filtration substantially decreased the amount of retained solvent,

which also reduced the amount of contaminant pigment species in agreement with this work.³³ These considerations highlight the potential of utilizing this contaminant removal framework to inform STRAP process design when considering the recovery of plastics from mixed plastic waste.

Conclusions

The removal of contaminants from plastic waste is a critical challenge for improving recycling efficiency and ensuring the safety of recycled materials. We have developed computational approaches for guiding solvent selection to facilitate contaminant removal. By leveraging molecular models, it is possible to identify suitable solvents that selectively remove target small-molecule contaminants and recover high-quality target polymer resins. We specifically consider the design of two contaminant removal methods based on leaching or on the integration of contaminant removal considerations during dissolution-based plastics recycling with the goal of achieving higher-purity recovered polymers. Two case studies leveraging either previously reported or newly collected experimental data validate the framework for the removal of brominated flame retardants and phthalates from common polymer resins. Two

additional case studies then apply the computational tools to identify solvents for the removal of a broad range of PFAS from LDPE and to determine the likely contaminant removal efficiency of steps in previously published plastics recycling processes.

Moving forward, further integration of computational modeling with experimental validation will enhance the predictive accuracy of solubility models and expand their applicability to a wider range of plastic waste streams. Other additive compounds in plastic products (such as stabilizers, plasticizers, modifiers, pigments, fillers, and processing aids) can also be taken into consideration with a similar approach. We note that the current methodology utilizes thermodynamic models to predict polymer solubility and contaminant removal at equilibrium and therefore neglects practical considerations, such as the kinetics of contaminant mass transfer,⁴³ which will be important to consider explicitly in future work. Because STRAP involves polymer-rich phases, contaminant partitioning may differ from the dilute-polymer limit modeled here. Our predictions therefore represent upper-bound removal efficiencies; future work will incorporate polymer-rich phase behavior and experimental validation. Similarly, we found that solvent retained within polymers recovered after STRAP would affect contaminant removal efficacy. The case studies on STRAP presented in this work use a single solvent for contaminant removal, with the same solvent leading to both polymer dissolution and precipitation depending on the temperature. However, polymer precipitation can alternatively be induced through the addition of an antisolvent,^{4,31} which then would lead to a solvent mixture that could further be used to tune contaminant removal strategies in future STRAP process designs. The computational approach also treats the partitioning of each contaminant separately, whereas interactions between contaminants may affect partitioning behavior. While preliminary analysis (described in the SI) indicates minimal such interference effects, further investigation of interactions between contaminants will be a subject of future work. Finally, we envision combining these methods with new computational tools, such as machine learning models, to aid in the selection of representative contaminant models in order to expand the scope of possible target contaminants considered during process design.

Conflicts of interest

There are no conflicts of interest to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information is available and includes one PDF file and one XLS file. The PDF contains computational methods for polymer solubility calculations, a summary of literature experimental methods for BFR extraction, analysis of potential effects of dissolved contaminants on partitioning, nine tables summarizing

predicted log *P* values and listing solvents/contaminants, and four figures showing contaminant chemical structures and COSMO-RS calculation results. The XLS file contains additional data for PFAS miscibility and log *P* calculations, as well as phthalate miscibility calculations. See DOI: <https://doi.org/10.1039/d5gc06059a>.

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